

The 2019 National Elections in Guinea-Bissau: A Quantitative Analysis of Religious and Ethnic Voting

Appendix and Errata

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A Appendix

A1 Unit of Analysis

Figure A1 shows the distribution of the number of registered voters within each electoral district (*Distrito Eleitoral: DE*). The number of registered voters in a single electoral district ranges from 4 to 3,968. The average number of registered voters allocated within a district is 380.35. Only in the foreign constituencies in Africa and Europe and in the capital area (*Sector Autónomo de Bissau: SAB*) are there electoral districts with more than 2,000 registered voters.

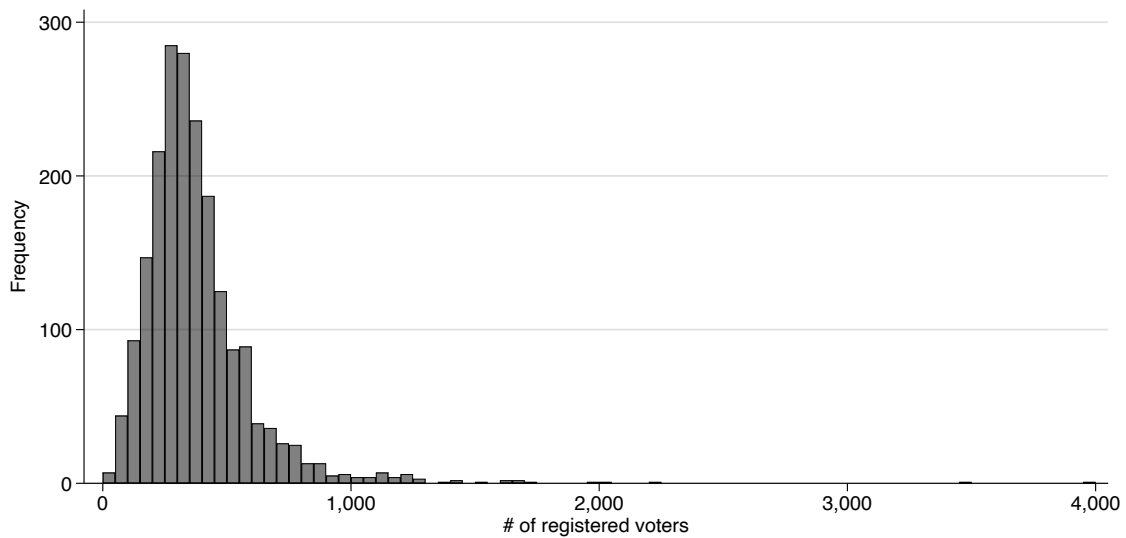


Figure A1: The distribution of the number of registered voters within each electoral district

A2 Descriptive Statistics

Table A1 presents the descriptive statistics of the variables used in the regression analysis. There are a total of 2,012 electoral districts for the 2019 national elections in Guinea-Bissau, but this table shows only 1,965 electoral districts because 47 districts are excluded from the analysis: 20 are the districts in foreign constituencies, 17 have missing data on the election results, and 10 have discrepancies in the number of registered voters between the voter registration data and the election results data, as explained in the main text.

Within the 1,965 electoral districts shown in Table A1, there are some cases where vote shares for some parties or candidates are more than one, as indicated by the maximum values of vote shares. This suggests that in some electoral districts the number of votes received by parties or candidates exceeds the number of registered voters. Therefore, I exclude from the analysis those electoral districts where the total number of votes is higher than the number of registered voters, probably due to invalid data on voter registration or election results. This explanation should also have been included in the main text, but it was inadvertently removed from the main text during the revision process.

Figures A2–A4 visualize the distribution of the three control variables by region. These figures are based on registered voters in all electoral districts, including those excluded from the analysis. Figure A2 shows the distribution of registered voters' age. Note that some registered voters were aged 140 or older according to the voter registration data. Figure A3 shows the distribution of their profession types. *NA* includes no answers and non-applicable answers (e.g., telephone numbers), while *Others* includes all professions other than housekeepers and farmers, such as teachers, technicians, and civil servants. Figure A4 shows the distribution of the types of documents used by voters to register.

Table A1: Descriptive statistics

Variables	N	Mean	Std. Dev.	Min.	Max.
Key explanatory variables					
Muslim rate	1,965	0.191	0.142	0.000	0.542
Fula rate	1,965	0.250	0.299	0.000	0.985
Control variables					
Average age	1,965	36.802	2.342	28.667	48.261
No profession rate	1,965	0.285	0.204	0.000	0.989
Housekeeper rate	1,965	0.310	0.166	0.000	0.771
Farmer rate	1,965	0.092	0.089	0.000	0.498
ID card rate for registration	1,965	0.350	0.243	0.000	0.893
Dependent variables					
MADDEM-G15 vote share	1,965	0.169	0.117	0.005	1.766
PAIGC vote share	1,965	0.260	0.162	0.013	1.957
Embaló vote share (round 1)	1,965	0.210	0.163	0.000	0.778
Pereira vote share (round 1)	1,965	0.277	0.158	0.000	1.750
Embaló vote share (round 2)	1,965	0.408	0.151	0.022	1.250
Pereira vote share (round 2)	1,965	0.322	0.158	0.021	1.750

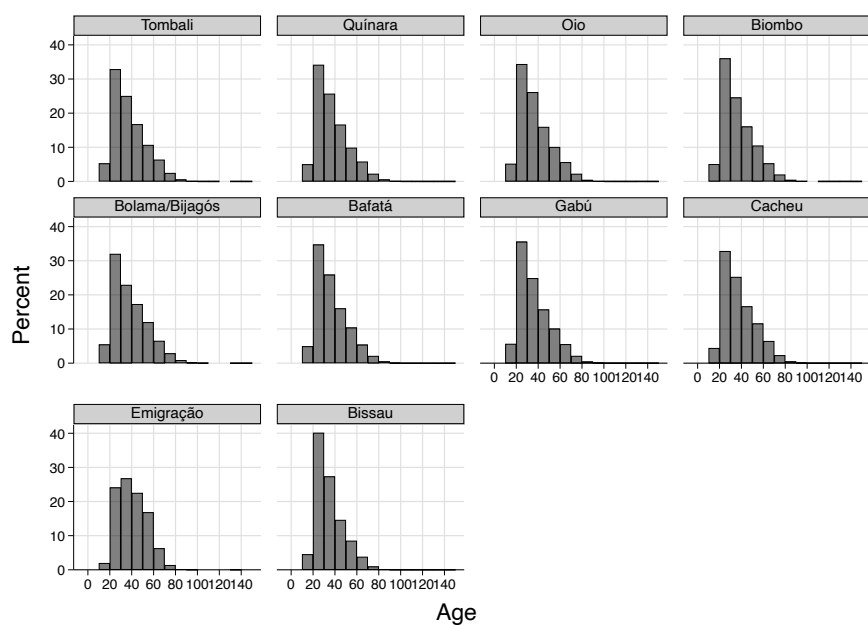


Figure A2: Age of registered voters by region

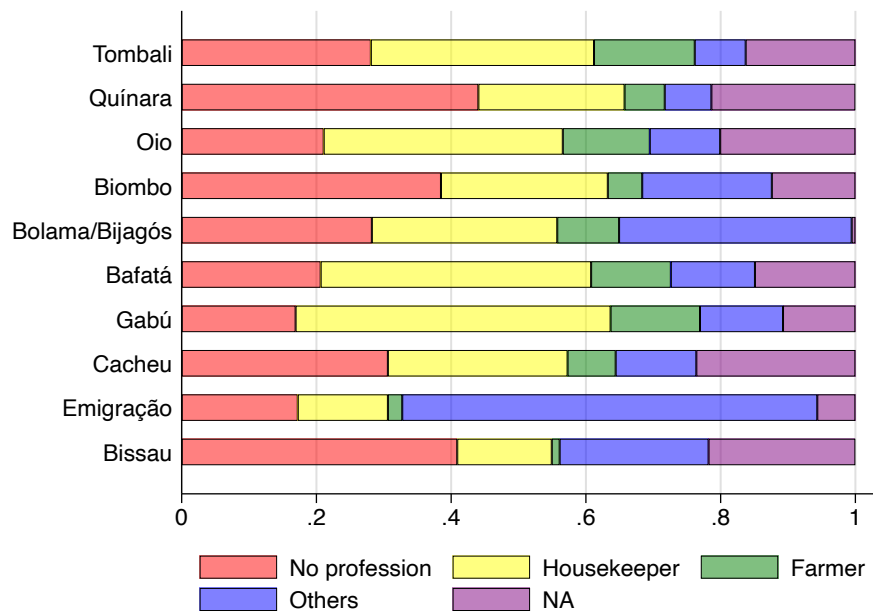


Figure A3: Profession types of registered voters by region

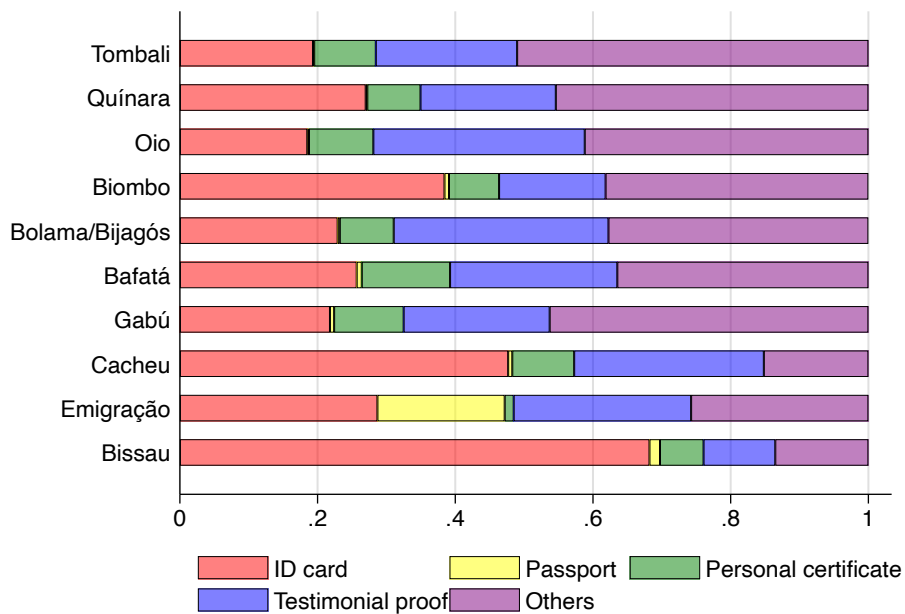


Figure A4: Types of documents that voters used for registration by region

A3 Correlations

Figures A5 and A6 illustrate the correlation between key explanatory variables and dependent variables. Figure A5 shows the correlations between the Muslim rate (i.e., the rate of registered voters with popular names of Arabic origin) and the vote share for the Movement for a Democratic Alternative, Group of 15 (*Movimento para a Alternância Democrática-Grupo dos 15*: MADEM-G15) and its presidential candidate Umaro Sisocco Embaló in the three national elections, respectively. Figure A6 shows the correlations between the Fula rate and the vote share for MADEM-G15 and Embaló. Each gray dot represents an electoral district used for the analysis, while each thick line is a linear prediction plot for the Muslim/Fula rate and the vote share.

According to Figure A5, the MADEM-G15 vote share had a clear and positive correlation with the Muslim rate in the legislative election. This suggests that MADEM-G15 was more popular in electoral districts with a higher proportion of Muslim residents. However, according to Figure A6, there does not appear to be a clear correlation between the Fula rate and MADEM-G15's vote share.¹ These findings suggest that MADEM-G15 did not gain significant popularity in electoral districts with a larger Fula population, even though it tended to receive more support from the Muslim community as a whole, including the Fula.

In contrast, in the presidential election, Figures A5 and A6 clearly show significant and positive correlations between the Muslim/Fula rate and the vote share for Embaló, especially in the first round. In the second round of the presidential election, Embaló's

¹The correlation coefficients between the Muslim/Fula rate and the vote shares for MADEM-G15/Embaló are positive and statistically significant at least at the 5% significance level in all panels of Figures A5 and A6, regardless of the strength of the correlation.

vote share increased even in electoral districts with fewer Muslim/Fula registered voters, and thus, the correlation becomes weaker compared to the first round.

This trend can be seen in the geographic distributions shown in Figure A7. In the maps, the darker the color, the higher the Muslim/Fula rates and vote shares for MADEM-G15 and Embaló. The sectors are color-coded on a scale from 0 to 1, with ten different levels.² The Muslim/Fula rates of registered voters are higher in the eastern sectors, such as those in the Bafatá and Gabú regions, and the vote share for Embaló in the first round was also higher in these eastern sectors. However, in the second round, his vote share became more evenly distributed throughout the entire country.

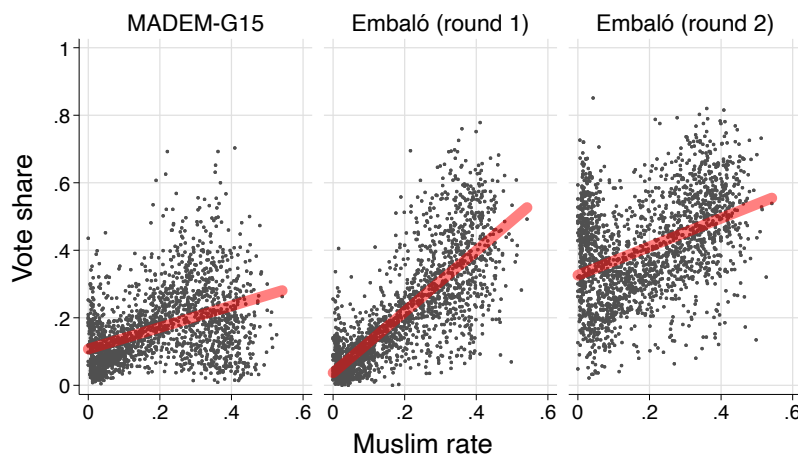


Figure A5: Muslim rate and vote shares for MADEM-G15 and Embaló

²Certain sectors, such as Komo, Ondame, Uno, Ingoré, and Calequisse, are considered separate sectors in the voter registration and election results data, but are considered part of other sectors in the map data. As a result, the sectors shown in Figure A7 do not necessarily correspond to the sectors indicated by the dummy variables incorporated into the estimation models.

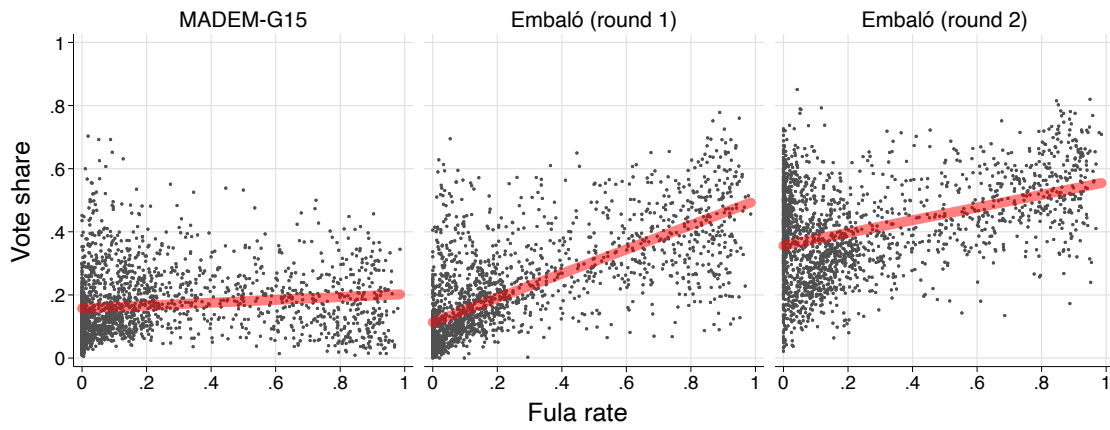


Figure A6: Fula rate and vote shares for MADEM-G15 and Embaló

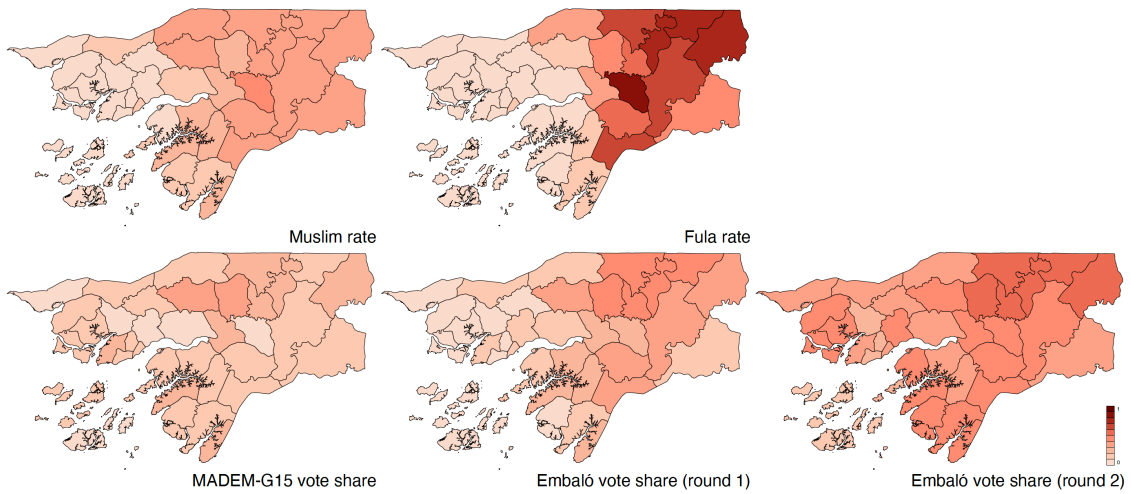


Figure A7: Geographic distributions of the Muslim/Fula rates of registered voters and vote shares for MADEM-G15 and Embaló

A4 Discussion

In this subsection, I will delve into the main results of the analysis, with an emphasis on two critical findings that do not directly align with my hypotheses, or expected outcomes based on the hypotheses, regarding the voting results for MADEM-G15 and Embaló.

The Decline of Religious and Ethnic Voting during the Presidential Election Rounds

One finding of this study is the diminishing impact of the Muslim/Fula rate from the first to the second round of the presidential election. Given that the second round determines the ultimate winner of the presidential election, it would be reasonable to expect MADEM-G15 to intensify its efforts to appeal to specific religious and ethnic groups, and for voters in the Muslim and Fula communities to become more motivated to support MADEM-G15 in this round. However, the regression results do not support this expectation.

The decline in the strength of religious and ethnic voting in the presidential election rounds could be attributed to the fact that Embaló was able to win political support across electoral districts, regardless of their religious and ethnic characteristics. The fact that Embaló increased his popularity even in electoral districts with a smaller proportion of Muslim/Fula voters in the second round, implies that his vote share was no longer so closely tied to the religious and ethnic characteristics of registered voters.

Figure A8 overlays two scatter plots to compare Embaló's strength against Pereira in two presidential election rounds. The vertical axis indicates an electoral district identification number, which is ordered by region from Tombali to Bissau. The horizontal axis indicates the ratio of votes for Embaló to votes for Pereira in natural logarithmic form. This indicator is negative when Embaló received fewer votes than Pereira in an electoral district, and

positive when Embaló received more. Figure A8 shows that the ratio increased in most electoral districts, except for those in the Bafatá and Gabú regions, where Embaló was likely to obtain more votes than Pereira, even in the first round. It indicates that, after the first round, Embaló managed to win votes throughout the country (especially in rural area, without Bissau), so that the impact of the Muslim and Fula rates decreased in the second round. His popularity throughout the country was probably due to the political support he picked up from other presidential candidates whose runs ended in the first round, such as Nuno Gomes Nabiam, José Mário Vaz, and Carlos Gomes Júnior.

The Negative Impact of the Fula Population in the Legislative Election

Another finding is that the Fula rate had a negative influence on the vote share for MADEM-G15 in the legislative election, as shown in Model 7 in Table 8.5. This unexpected impact of Fula voters may be due to the ethnicity of MADEM-G15's national coordinator.

MADEM-G15 was founded by 15 former members of the African Party for the Independence of Guinea and Cape Verde (*Partido Africano para a Independência da Guiné e Cabo Verde*: PAIGC), including Braima Camará, who belongs to the Mandinka ethnic group, and was elected national coordinator of MADEM-G15. Consequently, in the legislative election, MADEM-G15 was generally perceived as representing the Mandinka group, while its candidate in the presidential election, Embaló, was seen to represent the Fula group due to his ethnicity. This difference in ethnicity may explain why the regression results for the legislative election show no benefit to the MADEM-G15 vote share from the Fula community. This result contrasts with that for the presidential election, which shows that Embaló received more votes in districts where more Fula resided because of his

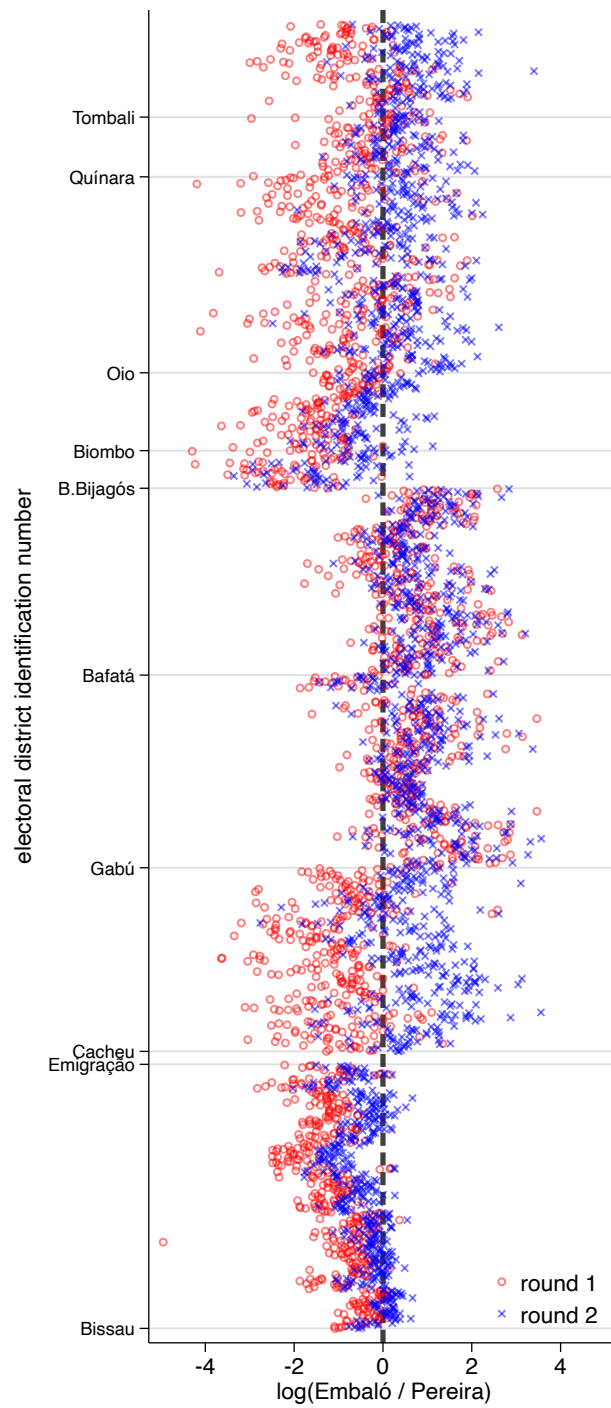


Figure A8: The ratio of Embaló's votes to Pereira's votes in two rounds of the presidential election

coethnic ties with the local population.

This result may also be attributed to the historical reluctance of the Fula population to vote for the Mandinka party, MADEM-G15. In the 19th century, armed forces from the Futa Jallon, an Islamic State dominated by the Fula, invaded the Kaabu Empire ruled by Mandinka. The Battle of Kansala, fought in the capital of the Kaabu Empire, led to its downfall and the beginning of Fula influence in the region. It is intriguing to observe how this historical confrontation between ethnic groups continues to shape contemporary political dynamics in Guinea-Bissau.³

³The people in the two ethnic groups, Fula and Mandinka, share the religious characteristics of Islam. As shown in the regression results, the Muslim rate has a consistent impact on increasing the vote share for MADEM-G15 and Embaló.

B Errata

In the first edition of *Political Instability in Guinea-Bissau*, Table 8.4 on page 155 and Table 8.5 on page 157 contained typographical errors due to an editorial oversight. A control variable *No profession rate* is statistically significant at the 10% level in Models 6 and 12, respectively, but the significance level was not correctly reported by asterisks. The tables should read as follows. Note that revisions are indicated in red text:

Table 8.4: OLS regression results on religious voting

	Model 1	Model 2	Model 3	Model 4	Model 5	Model 6
	MADDEM	Embaló Round 1	Embaló Round 2	PAIGC	Pereira Round 1	Pereira Round 2
Muslim rate	0.440*** (0.036)	0.744*** (0.031)	0.273*** (0.037)	0.166*** (0.030)	0.124*** (0.033)	-0.003 (0.036)
Average age	0.001 (0.001)	-0.001 (0.001)	-0.000 (0.002)	0.003** (0.001)	0.006*** (0.002)	0.006*** (0.002)
No profession rate	-0.041*** (0.015)	0.005 (0.015)	0.041** (0.019)	-0.025 (0.017)	-0.037** (0.018)	-0.034* (0.019)
Housekeeper rate	0.032 (0.030)	0.032 (0.026)	0.027 (0.030)	-0.021 (0.026)	-0.009 (0.029)	-0.006 (0.030)
Farmer rate	-0.229*** (0.049)	0.042 (0.044)	0.072 (0.055)	-0.095** (0.044)	-0.157*** (0.052)	-0.065 (0.054)
ID card rate for registration	0.012 (0.019)	-0.101*** (0.017)	-0.279*** (0.023)	0.237*** (0.020)	0.182*** (0.021)	0.153*** (0.023)
Constant	0.087* (0.045)	0.037 (0.038)	0.456*** (0.060)	0.041 (0.052)	-0.024 (0.058)	0.031 (0.063)
Sector dummy variables	YES	YES	YES	YES	YES	YES
<i>N</i>	1,926	1,958	1,959	1,926	1,958	1,959
<i>R</i> ²	0.410	0.763	0.526	0.664	0.595	0.538

Source: Voter registration list and election results data for the 2019 national elections in Guinea-Bissau.

Note: Robust standard errors in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table 8.5: OLS regression results on ethnic voting

	Model 7	Model 8	Model 9	Model 10	Model 11	Model 12
	MADDEM	Embaló Round 1	Embaló Round 2	PAIGC	Pereira Round 1	Pereira Round 2
Fula rate	-0.065*** (0.019)	0.195*** (0.019)	0.102*** (0.017)	0.042*** (0.014)	-0.005 (0.016)	-0.059*** (0.017)
Average age	-0.002 (0.001)	-0.002** (0.001)	-0.000 (0.002)	0.002* (0.001)	0.005*** (0.002)	0.005*** (0.002)
No profession rate	-0.048*** (0.016)	-0.017 (0.016)	0.032* (0.019)	-0.030* (0.017)	-0.039** (0.018)	-0.033* (0.019)
Housekeeper rate	0.063** (0.030)	0.068** (0.030)	0.039 (0.031)	-0.012 (0.026)	-0.002 (0.029)	-0.004 (0.030)
Farmer rate	-0.326*** (0.054)	-0.140*** (0.052)	0.002 (0.056)	-0.137*** (0.043)	-0.184*** (0.052)	-0.059 (0.053)
ID card rate for registration	0.012 (0.020)	-0.085*** (0.019)	-0.272*** (0.023)	0.241*** (0.020)	0.183*** (0.022)	0.151*** (0.023)
Constant	0.270*** (0.049)	0.185*** (0.042)	0.495*** (0.059)	0.074 (0.051)	0.018 (0.058)	0.059 (0.062)
Sector dummy variables	YES	YES	YES	YES	YES	YES
<i>N</i>	1,926	1,958	1,959	1,926	1,958	1,959
<i>R</i> ²	0.344	0.697	0.521	0.660	0.592	0.541

Source: Voter registration list and election results data for the 2019 national elections in Guinea-Bissau.

Note: Robust standard errors in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.